

Data Interpretation Exercise

Data Interpretation Assessment (R Programming)

Topics:

- Mapping Data onto Aesthetics
- Coordinate Systems and Axes
- Color Scales

Instructions: Use the given dataset to write an appropriate R program, generate the required visualization, and answer the questions.

Question 1: Mapping Data onto Aesthetics

Dataset

Product	Sales (₹ Lakhs)	Profit (%)	Category
P1	120	18	Electronics
P2	90	12	Clothing
P3	150	25	Electronics
P4	80	8	Furniture
P5	135	22	Furniture
P6	100	16	Clothing

Question

1. Write an R program to create a scatter plot with **Sales** on the x-axis and **Profit** on the y-axis.

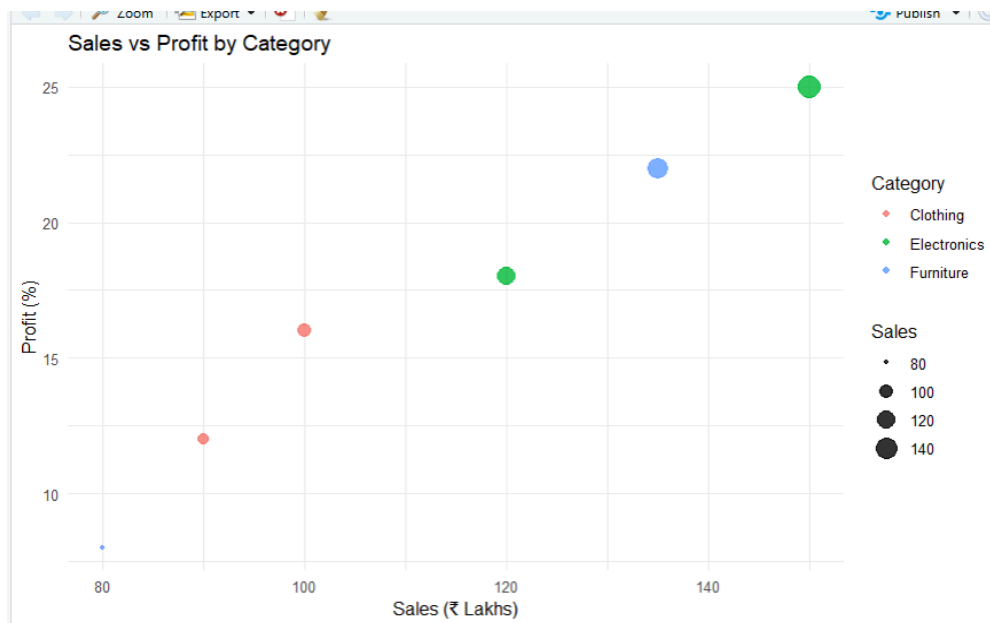
Answer:

```
.libPaths("D:/Rlibrary")
library(ggplot2)

data <- data.frame(
  Product = c("P1", "P2", "P3", "P4", "P5", "P6"),
  Sales = c(120, 90, 150, 80, 135, 100),
  Profit = c(18, 12, 25, 8, 22, 16),
```

```
Category = c("Electronics", "Clothing", "Electronics", "Furniture", "Furniture",
  "Clothing")
)
```

```
ggplot(data, aes(x = Sales, y = Profit, color = Category, size = Sales)) +
  geom_point(alpha = 0.8) +
  labs(
    title = "Sales vs Profit by Category",
    x = "Sales (₹ Lakhs)",
    y = "Profit (%)",
    color = "Category",
    size = "Sales"
  ) +
  theme_minimal()
```



2. Map **Category** using an appropriate aesthetic and **Sales** using another suitable aesthetic.

Answer:

The **Category** variable is mapped using the color aesthetic, while the **Sales** variable is mapped using the size aesthetic to make the visualization more informative.

3. Interpret the relationship between Sales and Profit.

Answer:

The scatter plot indicates a **positive relationship** between Sales and Profit, showing that products with higher sales generally have higher profit percentages.

4. Identify the type of scale used for each variable.

Answer:

VARIABLE	SCALE
Product	Nominal
Sales	Continuous
Profit	Continuous
Category	Nominal

5. Suggest one alternative visualization that better represents the same data.

Answer:

A **Bubble Chart** is a suitable alternative because it can represent Sales on the x-axis, Profit on the y-axis, Category using color, and Sales using bubble size.

Question 2: Coordinate Systems (Cartesian and Polar)

Dataset

Direction	Frequency
N	18
NE	12
E	20
SE	15
S	22

SW W NW	11
	16
	14

Questions

1. Write an R program to visualize the dataset using a Cartesian coordinate system.

Answer:

```
Direction <- c("N","NE","E","SE","S","SW","W","NW")
```

```
Frequency <- c(18,12,20,15,22,11,16,14)
```

```
barplot(Frequency,
```

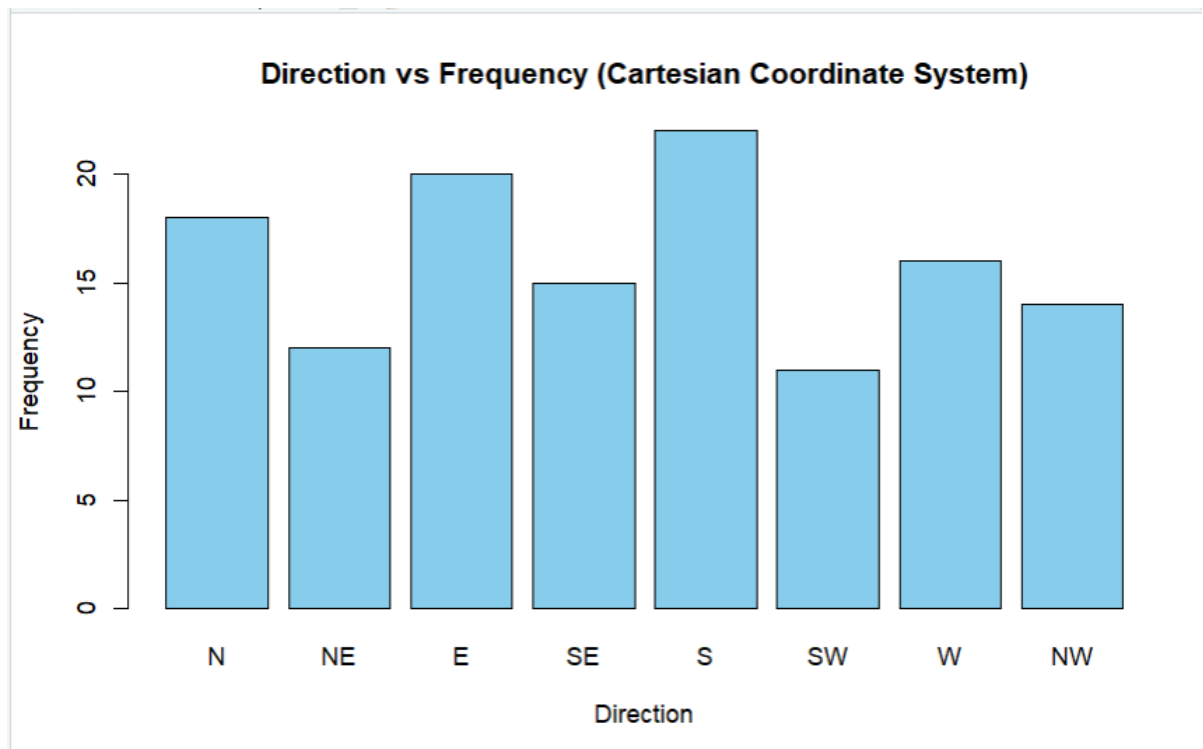
```
  names.arg = Direction,
```

```
  col = "skyblue",
```

```
  main = "Direction vs Frequency (Cartesian Coordinate System)",
```

```
  xlab = "Direction",
```

```
  ylab = "Frequency")
```



2. Write another R program to visualize the same data using a Polar coordinate system.

Answer:

```
Direction <- c("N","NE","E","SE","S","SW","W","NW")
```

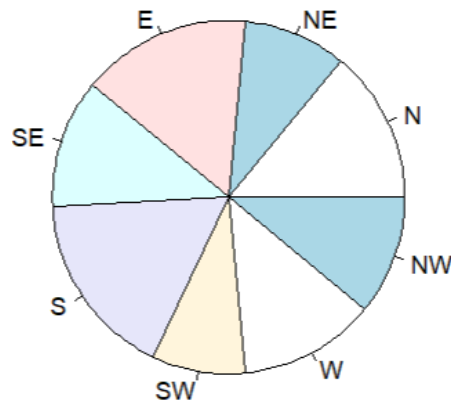
```
Frequency <- c(18,12,20,15,22,11,16,14)
```

```
pie(Frequency,
```

```
  labels = Direction,
```

```
  main = "Direction vs Frequency (Polar Coordinate System)")
```

Direction vs Frequency (Polar Coordinate System)



3. Compare the effectiveness of both visualizations.

Answer:

The Cartesian coordinate system is effective for comparing the frequency values of different directions because the bar heights can be easily compared. In contrast, the Polar coordinate system emphasizes the circular arrangement of directions, making it easier to understand the overall directional distribution.

4. Explain which coordinate system is more appropriate for directional data.

Answer:

The Polar coordinate system is more appropriate for directional data because it represents directions as angles around a circle, which closely matches the way compass directions are organized. This makes it easier to identify directional patterns and relationships in the dataset.

5. Justify your answer based on data interpretation.

Answer:

The Polar coordinate system provides a more meaningful representation of directional data by displaying all directions in their natural circular order. This helps in interpreting the frequency of each direction more effectively and makes the overall directional pattern easier to understand than the Cartesian coordinate system.

Question 3: Color Scales for Data Representation

Dataset

Student	Marks
A	45
B	58
C	72

Student	Marks
D	81
E	67
F	90

Questions

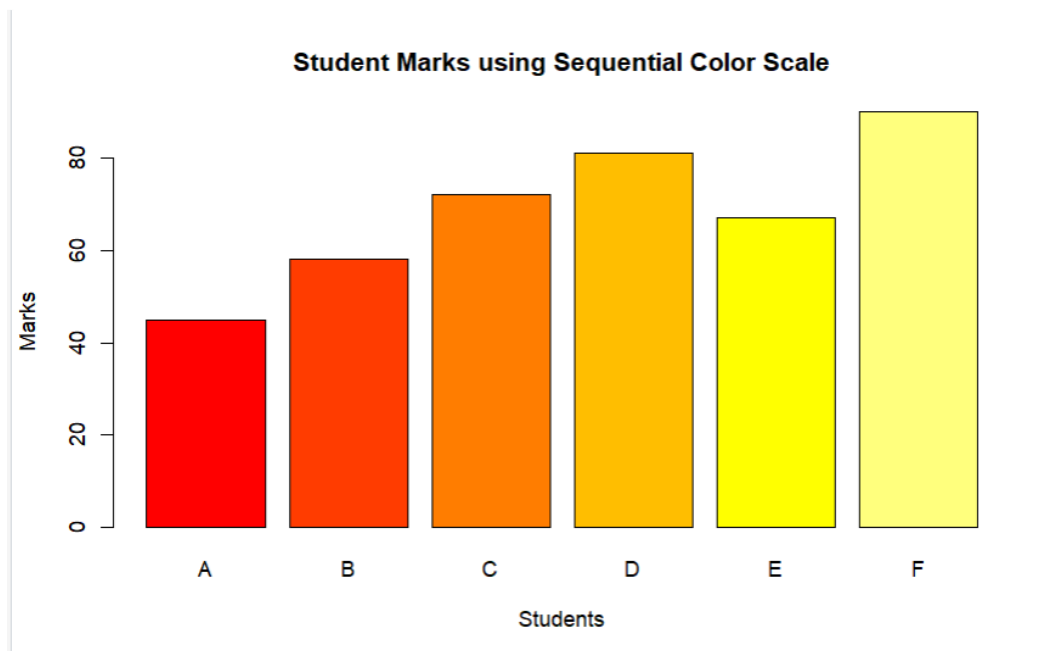
1. Write an R program to create a bar chart using a sequential color scale based on Marks.

```
Student <- c("A", "B", "C", "D", "E", "F")
```

```
Marks <- c(45, 58, 72, 81, 67, 90)
```

```
colors <- heat.colors(length(Marks))
```

```
barplot(Marks,  
        names.arg = Student,  
        col = colors,  
        main = "Student Marks using Sequential Color Scale",  
        xlab = "Students",  
        ylab = "Marks")
```



2. Identify the students with the highest and lowest marks from the visualization.

Answer:

From the visualization, **Student F** has the highest marks with **90**, while **Student A** has the lowest marks with **45**. The bar heights and color intensity make these values easy to identify.

3. Explain why a sequential color scale is appropriate.

Answer:

A sequential color scale is appropriate because the marks are continuous numerical values that range from low to high. The gradual change in color intensity helps viewers easily distinguish lower marks from higher marks and understand the data distribution.

4. Suggest a color palette suitable for color-blind users.

Answer:

A color-blind-friendly palette such as **Viridis** is a suitable choice because it provides clear color differences while remaining easily distinguishable for most users with color vision deficiencies.

5. Explain the difference between sequential and categorical color scales.

Answer:

SEQUENTIAL COLOR SCALES	CATEGORICAL COLOR SCALES
<i>It is used to represent continuous numerical data with an ordered range.</i>	<i>It is used to represent distinct categories without any natural order.</i>
<i>The color changes gradually from light to dark or from one shade to another.</i>	<i>Each category is assigned a different and distinct color.</i>
<i>It helps show the magnitude or intensity of values.</i>	<i>It helps distinguish different groups or categories.</i>
<i>Example: Marks, temperature, or sales values.</i>	<i>Example: Student groups, product categories, or departments.</i>

Question 4: Highlighting with Color

Dataset

Region	Revenue (₹ Crores)
North	820
South	760
East	790
West	1010
Central	770

Questions

1. Write an R program to create a bar chart highlighting the region with the highest revenue.

```
Region <- c("North", "South", "East", "West", "Central")
```

```
Revenue <- c(820, 760, 790, 1010, 770)
```

```
colors <- c("lightblue", "lightblue", "lightblue", "red", "lightblue")
```

```
barplot(Revenue,
        names.arg = Region,
        col = colors,
        main = "Revenue by Region",
        xlab = "Region",
        ylab = "Revenue (₹ Crores)")
```



2. Explain how highlighting improves data interpretation.

Answer:

Highlighting improves data interpretation by drawing attention to the most important data point, allowing viewers to quickly identify significant values without examining every bar in the chart.

3. Identify the highest-performing region.

Answer:

The **West** region is the highest-performing region with a revenue of **₹1010 crores**, making it the most prominent bar in the visualization.

4. Discuss how excessive use of colors can affect interpretation.

Answer:

Using too many different colors can make the visualization confusing and distract viewers from the important information. It may also reduce readability and make it difficult to identify the highlighted data.

5. Suggest one alternative highlighting technique other than color.

Answer:

One alternative highlighting technique is to add data labels or annotations to the important bar. This helps emphasize the highest value without relying only on color, making the visualization easier to understand.

Question 5: Integrated Visualization

Dataset

City	Population (Million)	AQI	Rainfall (mm)	Region
A	8.2	110	850	North
B	5.6	180	430	South
C	9.4	90	1200	North
D	6.8	240	520	East
E	4.9	150	680	West

Questions

1. Write an R program to create a bubble chart using appropriate aesthetic mappings.

```
.libPaths("D:/Rlibrary")
```

```
library(ggplot2)
```

```
City <- c("A", "B", "C", "D", "E")
```

```
Population <- c(8.2, 5.6, 9.4, 6.8, 4.9)
```

```
AQI <- c(110, 180, 90, 240, 150)
```

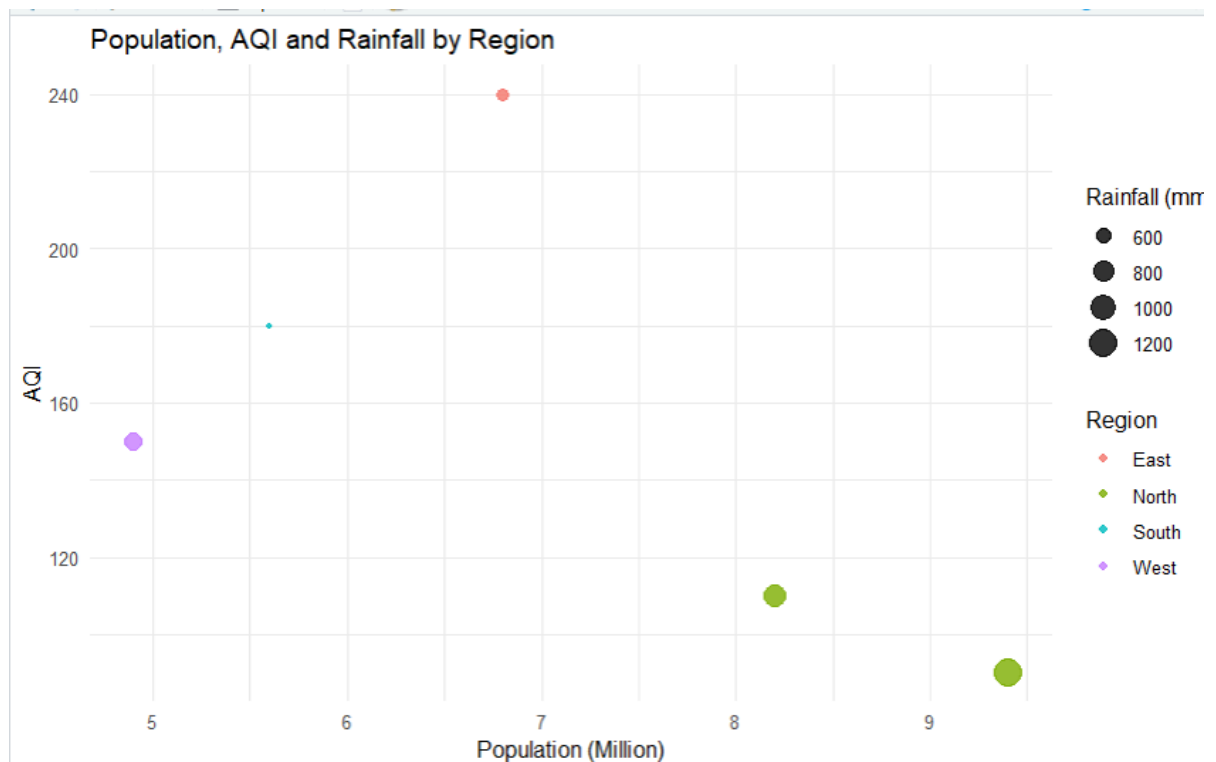
```
Rainfall <- c(850, 430, 1200, 520, 680)
```

```
Region <- c("North", "South", "North", "East", "West")
```

```
data <- data.frame(City, Population, AQI, Rainfall, Region)
```

```
ggplot(data, aes(x = Population, y = AQI, size = Rainfall, color = Region)) +
```

```
geom_point(alpha = 0.8) +
labs(
  title = "Population, AQI and Rainfall by Region",
  x = "Population (Million)",
  y = "AQI",
  size = "Rainfall (mm)",
  color = "Region"
) +
theme_minimal()
```



2. Select suitable aesthetics for Population, AQI, Rainfall, and Region.

Answer:

The **Population** is represented on the **x-axis**, **AQI** is represented on the **y-axis**, **Rainfall** is represented using the **bubble size**, and **Region** is represented using different **colors**.

3. Interpret the relationship between Population and AQI.

Answer:

The bubble chart shows that there is **no strong positive relationship** between Population and AQI. Some cities with higher populations have lower AQI values, while others with lower populations have higher AQI values, indicating that AQI is influenced by factors other than population alone.

4. Identify the continuous and categorical variables.

CONTINUOUS VARIABLES	CATEGORICAL VARIABLES
Population	City
AQI	Region
Rainfall	—

5. Recommend one improvement to enhance the visualization's readability.

Answer:

The readability of the visualization can be improved by adding **data labels** to each bubble so that viewers can easily identify the corresponding city without referring to the legend repeatedly.